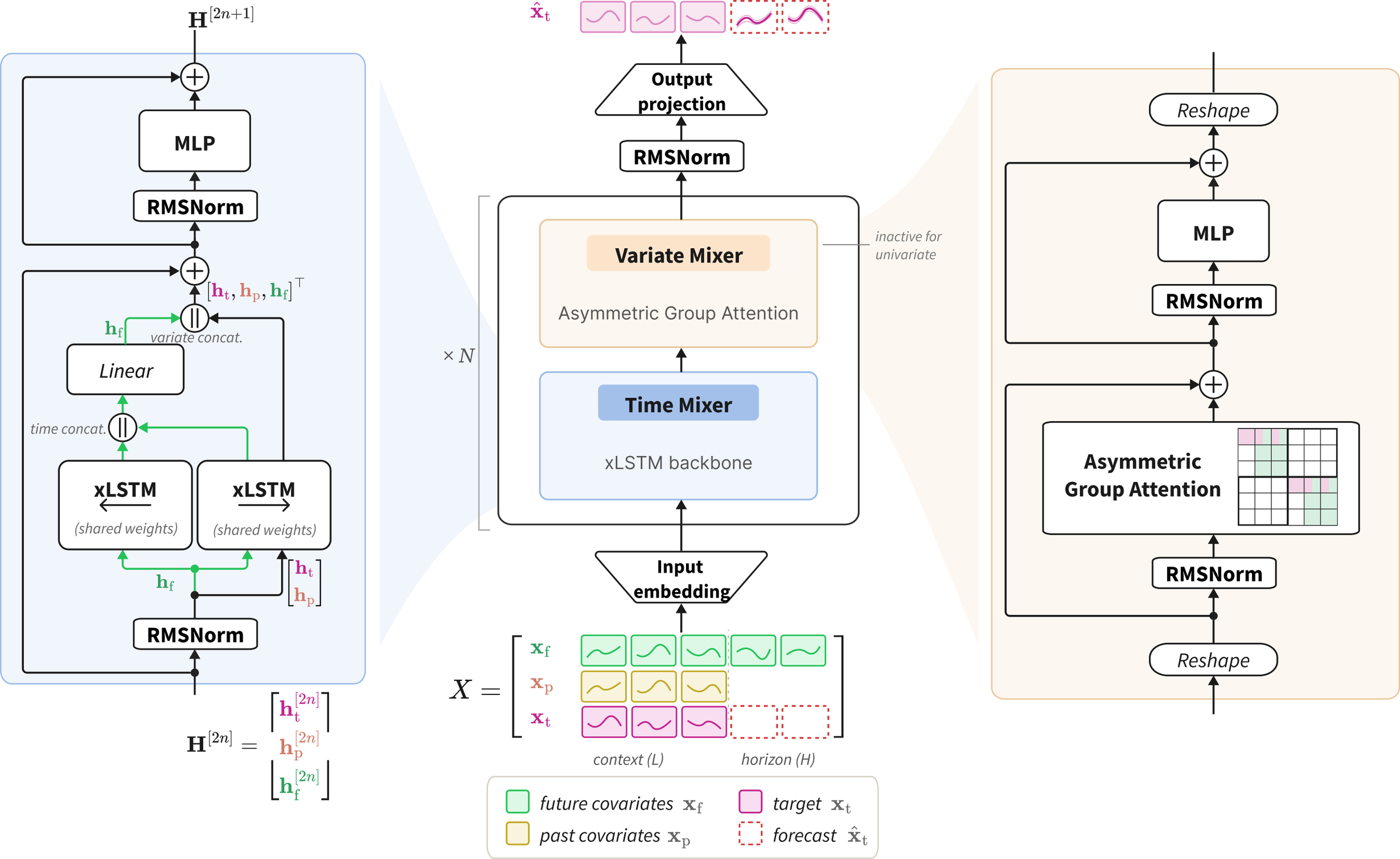

$\hat{\mathbf{x}}_t$
Output projection
RMSNorm
Variate Mixer
Asymmetric Group Attention
inactive for univariate
Time Mixer
xLSTM backbone
$\times N$
Input embedding
$X = \begin{bmatrix} \mathbf{x}_f \\ \mathbf{x}_p \\ \mathbf{x}_t \end{bmatrix}$
context (L)
horizon (H)
future covariates $\mathbf{x}_f$
target $\mathbf{x}_t$
past covariates $\mathbf{x}_p$
forecast $\hat{\mathbf{x}}_t$
$\mathbf{H}^{[2n+1]}$
MLP
RMSNorm
$[\mathbf{h}_t, \mathbf{h}_p, \mathbf{h}_f]^\top$
$\mathbf{h}_f$
variate concat.
Linear
time concat.
xLSTM
(shared weights)
xLSTM
(shared weights)
$\mathbf{h}_f$
$\begin{bmatrix} \mathbf{h}_t \\ \mathbf{h}_p \end{bmatrix}$
RMSNorm
$\mathbf{H}^{[2n]} = \begin{bmatrix} \mathbf{h}_t^{[2n]} \\ \mathbf{h}_p^{[2n]} \\ \mathbf{h}_f^{[2n]} \end{bmatrix}$
Reshape
MLP
RMSNorm
Asymmetric Group Attention
RMSNorm
Reshape